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AI-Driven Traffic Accident Prediction: Integrating Real-Time Weather, Analytics, and Machine Learning for Enhanced Road Safety

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Abstract*— Accidents on the road continue to cause injuries and deaths globally, which represents one of the key factors affecting the safety of the population and city management. Conventional methods of accident prevention may be based on past experiences or even the passive measurement of risks and not on the dynamic environments. The study proposes a traffic accident forecasting system based on an artificial intelligence approach to predict traffic accident-prone areas using input data on real-time weather condition, traffic jam, and historical data on accidents to make actionable recommendations to promote road safety. The system employs the use of sophisticated machine learning models to combine characteristics of the environment with infrastructure factors to achieve probabilistic accident predictions with an associated confidence score. Weather conditions such as temperature, precipitation, visibility and wind resistance are accessed from the Open-Meteo API as real-time, and mapping functions and geolocation services come through the OpenStreetMap and geocoding services, respectively. Data analytics and visualization such as heatmaps and time-temporal trend analysis provide complete data information that allows stakeholders to identify spatial and time trends in the occurrence of accidents. Dynamic risk notifications and other informed strategies of traffic management are supported by the predictive model and thus can decrease the likelihood of accidents and enhance response planning.

Keywords*— Traffic Accidents, Road Safety, Artificial Intelligence, Machine Learning, Accident Prediction, Predictive Analytics, Real-Time Weather Data, Traffic Conditions, Geolocation-Based Risk Assessment SDG Goals -- SDG 3 – Good Health and Well-being SDG 9 – Industry, Innovation and Infrastructure, SDG 11 – Sustainable Cities and Communities

I. INTRODUCTION

The number of road traffic accidents worldwide remains one of the biggest problems in the world leading to a considerable number of injuries and deaths, as well as economic losses. Although the field of urban planning and traffic management has improved, traditional approaches to preventing accidents tend to be based on stagnant or historical data, recording or secondary prevention profiles, which are not enough when it comes to dynamic conditions. The study will touch upon the development of AI-assisted traffic accident prediction service that takes into consideration real-time environmental and infrastructural data to foresee high-risk locations and give recommendations that can be addressed to improve road safety. The system aims to minimize occurrence of accidents and to enhance the process of emergency response planning through the application of machine learning and predictive analytics.

The proposed system takes as input many other sources of data: real-time weather, amount of traffic congestion, and historical accident rates in an area, to model probabilistic estimations of accident hotspots. The Open-Meteo API is used to retrieve weather parameters (temperature, precipitation, visibility, and wind speed), whereas geolocation and mapping are covered with the use of OpenStreetMap and geocoding services. These multifactorial inputs are analyzed using machine learning models to come up with a calculation of accident risk scores and their levels of confidence. Heatmap and the graph of general temporal trends would act as the visualization tools to help the stakeholders determine the spatial and time oriented patterns, which would be helpful in making the evidence-based decisions regarding traffic patterns management and urban safety policy.

The inspiration in carrying out this research is based on the growing demands to have smart and real-time interventions in traffic management. The traditional systems have poor capability to react dynamically to dynamic weather conditions, traffic or environmental conditions. Through the predictive analytics integrated with the use of artificial intelligence, the system will be able to issue timely warnings, risk alerts, and trend analysis to allow the company to take proactive actions to preempt any accident. The study underlines the necessity to integrate the technological innovation and the goals of ensuring the safety of persons in the society in order to establish more robust and intelligent transportation systems.

The aims of the study are as follows: (1) to create a powerful machine learning-powered system capable of

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predicting traffic accidents based on both real-time and historical data; (2) to analyse and visualise the patterns of accidents through spatial and temporal analytics; (3) to evaluate risk factors of accidents such as weather and traffic conditions; (4) to improve decision-making by traffic authorities based on such predictive information; and (5) to create a scalable framework that may be used in various projects in urban planning, emergency response, and road safety improvement, among others.

Accuracy is enhanced through multi-source data fusion, feature-level normalization, supervised learning with cross-validation, and continuous model updating using user feedback.

II. Related Works

Shin and Yang [1] proposed a machine learning-based approach for predicting maritime accidents in Busan port by integrating historical maritime accident reports with Vessel Traffic Service (VTS) data. Their study focused on analyzing key operational parameters and environmental factors to identify high-risk zones and improve port safety management.

Ajah, Akanji, and Gomes [2] conducted a comprehensive review of recent trends in nuclear accident prediction. They evaluated different predictive models, highlighting the role of machine learning and statistical techniques in forecasting accident events and assessing safety risks in nuclear facilities.

Pandey et al. [3] developed a multi-model framework for accident prediction and risk assessment in steel operations. Their system combined simulation-based risk evaluation with machine learning algorithms to provide probabilistic predictions and enhance operational safety.

Zheng et al. [4] applied interpretable machine learning methods to predict Pressurized Water Reactor (PWR) accidents. Their approach emphasized quantitative parameter importance and operational thresholds, providing insights into critical factors that contribute to accident probability.

Liu et al. [5] introduced CCAF-Net, a cascade complementarity-aware fusion network for traffic accident prediction using dashcam video data. Their model leveraged multi-modal inputs and deep learning techniques to improve prediction accuracy in real-world traffic scenarios.

Tang et al. [6] developed a traffic accident severity level prediction model using improved machine learning algorithms. The study focused on assessing accident severity based on traffic, environmental, and infrastructural features to aid in emergency response planning.

Sajadi et al. [7] proposed an accident impact prediction model using a combination of deep convolutional and recurrent neural networks. Their approach captured both spatial and temporal patterns of accidents to accurately forecast accident consequences in urban settings.

Shintani et al. [8] developed a marine accident probability prediction model for pleasure boats using a regional ship accident database. Their system employed statistical and machine learning techniques to estimate the likelihood of accidents and support preventive measures.

Pei, Wen, and Pan [9] focused on traffic accident severity prediction through an interpretable deep learning model. Their research highlighted the importance of explainable AI in providing both accurate predictions and understandable insights for traffic management authorities.

Li et al. [10] proposed STGEN, a spatio-temporal generalized aggregation network for traffic accident prediction. The network effectively captured spatial and temporal dependencies in traffic data, improving the precision of accident forecasting models.

Behboudi, Moosavi, and Ramnath [11] reviewed recent advances in traffic accident analysis and prediction. Their study summarized various machine learning techniques, data sources, and challenges, providing a comprehensive overview of state-of-the-art methodologies.

Chai et al. [12] explored the use of non-visual data in enhancing road safety through machine learning-based accident prediction. Their work emphasized the integration of environmental, traffic, and behavioral data for improved predictive performance.

Wang et al. [13] introduced DeepAccident, a benchmark for motion and accident prediction in V2X autonomous driving environments. The study provided a dataset and evaluation framework for developing real-time predictive models in connected vehicle systems.

Lacherre, Castillo-Sequera, and Mauricio [14] conducted a systematic review of driving behavior factors influencing vehicle accident risk. Their research analyzed predictive models and highlighted explainability techniques for understanding risk factors in driver behavior.

Brandt et al. [15] developed a maritime accident risk prediction model integrating weather data with machine learning algorithms. Their approach focused on improving port and shipping safety by considering environmental conditions in accident forecasting.

Munim et al. [16] used automated machine learning to predict maritime accident risk. Their system optimized model selection and hyperparameters automatically, demonstrating improved prediction accuracy and reduced human intervention.

Wang et al. [17] addressed data imbalance in traffic accident prediction using adaptive graph structures and self-supervised learning. Their methodology improved predictive performance in datasets with rare but severe accident events.

Li et al. [18] proposed an interpretable traffic accident prediction approach using attention-based spatial-temporal multi-graph learning. The model captured complex traffic stream patterns while maintaining explainability for decision-makers.

Liu, Shen, and Gao [19] applied back-propagation neural networks for flight accident prediction. Their study focused on analyzing aviation safety data to identify high-risk conditions and enhance preventive measures.

Cui et al. [20] developed a sparse spatio-temporal dynamic learning approach for urban traffic accident forecasting. The model effectively handled large-scale traffic data and provided accurate predictions for proactive urban traffic management.

III. Existing and Proposed Methodology

The historical analysis, statistical trend analysis, and reactive reporting that are used as the main functioning methods of traditional traffic accident analysis can hardly fit the conditions in which the environment or traffic is subject to rapid changes. Available systems are currently based on the

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mere aggregation of prior accidents, manual risk ratings or fixed transportation models and thus lack in their capability to offer real-time, actionable intelligence. Instead, the suggested methodology takes advantage of artificial intelligence and LMs to combine dynamic sources of data, such as live weather forecasts, the latest information on traffic conditions, and geospatial data. This method enables the system to make probabilistic predictions with confidence scores, show accident hotspots using interactive maps and make time series trend analysis.

Table 1. Comparison of Previous Systems and Proposed System

Feature / Aspect	Previous Methodology	Proposed Methodology
Risk Assessment	No	Yes
Real-Time Adaptability	No	Yes
Visualization	No	Yes
Scoring algorithm	No	Yes
Predictive Capability	No	Yes
User Interaction	No	Yes
Decision Support	No	Yes

The table 1 gives a comparison of a traditional method of traffic accident analysis and the new AI-driven system. Unlike the existing approaches that still mostly rely on past data and manual evaluation of risks, the presented system incorporates the real-time weather, traffic, and geolocation information. It has predictive analytics, interactive visualizations, user feedback, which allows proactive measures to be made in preventing accidents, and can be used to make decisions on traffic management and safety promotion.

Conventional measures primarily concentrate on retrospective study of past records of accidents and are based on manual/statistical risk rating. They provide poor visualization, noninteractive style, and they cannot be made to adjust with changing conditions in real time. On the contrary, the suggested system utilizes a broad set of dynamic data sources such as a live weather forecast, traffic situations, and geolocation data and can predict accidents better and in a timely manner. Machine learning algorithms include calculations of probabilistic risk scores, and understanding of the accident-prone area can be acquired using interactive maps, heatmap, and trends analysis of the time-period. Also, individuals can use the system to report any incidents and this will form a feedback loop resulting in consistent accuracy of predictions. On balance, the methodology proposed changes the reactive practice into proactive allowing the traffic control and urban planners to take action in advance, deliver announcements

IV. Framework

The offered traffic accident prediction system (Figure 1) is based on a multi-step, structured approach to deliver insights that can be further used. The initial stage is the gathering of the data that consists of the historical accident records, real-time weather data by Open-Meteo API, traffic congestions, and geolocation details by OpenStreetMap and geocoding services.

Second step involves preprocessing and feature engineering of the data, cleaning, normalizing and transforming raw data to conventional features including temperature, precipitation, visibility, traffic density, road type and accident history. The third step is the development of the machine learning model when supervised learning algorithms process the multifactorial dataset and compute the probabilistic predictions of accident incidences and produce confidence scores. In step four, visualization and analytics are employed to generate interactive heat maps, spatial clustering, and time trend analysis to make the stakeholders aware of the accident-prone locations and high-risk periods of time. The fifth step is that the system allows the user to interact with the system and provide feedback, and thus, the users could report accidents that could be used to validate and continually improve the predictive models.

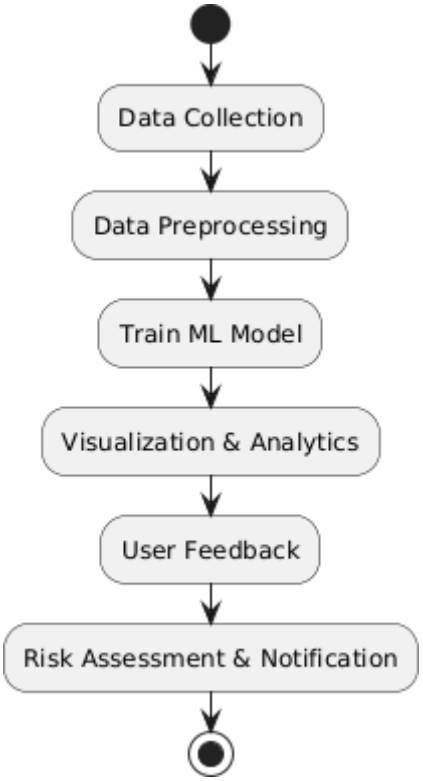


Fig 1. Working Flowchart

Step 1: Collection of Data

Here the big data is accumulated by various sources in order to introduce the main body of the predictive system. Past records on traffic accidents are recorded in order to establish previous trends and patterns of risk. Environmental conditions are captured using real-time data on the temperature, precipitation, wind speed, and visibility obtained by the Open-Meteo API. The data obtained using the geolocation services, such as OpenStreetMap, include the traffic congestion and road type data, while GPS coordinates directly identify the places prone to accidents.

Step 2: Data Cleaning and Data Transformation

Raw data preprocessing includes missing value imputation using mean and mode strategies, removal of duplicate accident records, normalization of numerical features using min–max scaling, and categorical encoding of road types and weather conditions. Feature transformation converts raw inputs into risk-aware variables such as weather severity index, traffic density score, and historical accident frequency. Temporal aggregation was applied to transform timestamp data into peak-hour indicators, weekday/weekend labels, and seasonal patterns.

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Step 3: Development of Machine Learning Model
Trained supervised machine learning algorithms on engineered and pre-processed dataset. In the model, relationships among environmental, traffic and historical accident factors are learned to perform probabilistic predictions. There is a measurement of the confidence of the output attached to each prediction Multiple supervised learning models including Random Forest, Gradient Boosting, and Logistic Regression were evaluated. Hyperparameter tuning was performed using grid search, and the final model was selected based on accuracy, precision, recall, and F1-score

Accuracy Enhancement Strategies –
Accuracy is enhanced using multiple strategies: (i) integration of heterogeneous data sources (weather, traffic density, road type, and historical accident data), (ii) feature engineering such as categorical encoding of road types and weather risk indexing, (iii) normalization and scaling to avoid feature dominance, (iv) model validation using k-fold cross-validation, and (v) continuous retraining using real-time user-reported accident feedback.

Step 4: Analytics and visualization
It provides the system to display accident patterns with interactive dashboards, spatial heatmaps, and temporal trend charts. The step will allow the stakeholders to single out the high-risk areas, high accident hours, and relationships with other risk elements. Visual representations are intuitive, which brings predictive insights to the fore.

Step 5: Interaction and Feedback by user

Customers are able to report real-life accidents and add information on the incidence which may include weather and traffic conditions at that particular time. This feedback can be used to validate the predictive models and overtime to make the determining process more accurate thus forming a continuous learning loop.

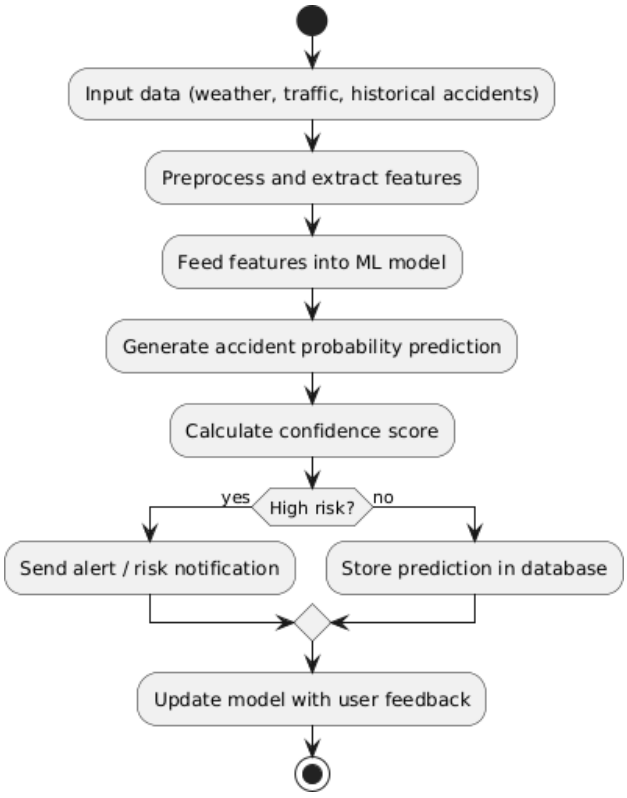


Fig 2. Algorithm workflow

The algorithm workflow (Figure 2) illustrates the process of predicting traffic accidents using the proposed AI system. It begins with collecting input data, including real-time weather conditions, traffic information, and historical accident records. The data is pre-processed and relevant features are extracted before being fed into the machine learning model. The model generates a probabilistic prediction of accidents along with a confidence score. Based on the predicted risk level, the system either sends alerts and notifications for high-risk situations or stores the prediction for future analysis. Additionally, user feedback from reported accidents is incorporated to continuously refine and improve the model's accuracy, ensuring dynamic adaptation to changing conditions.

Tree-based ensemble models were preferred due to their robustness to non-linear relationships and heterogeneous feature types. Random Forest demonstrated superior performance in terms of accuracy and precision compared to baseline logistic regression, making it suitable for accident risk prediction.

V. Results And Discussion

The given AI implemented traffic accident prediction model was tested on the base of the historical accident statistics as well as real-time weather conditions and traffic. Machine learning model proved to be highly accurate in assigning crash susceptible regions and the confidence scores indicated the confidence level with respect to different environmental and traffic conditions. Spatial heat maps were used to detect the existence of space clusters in highly hazardous areas and time analysis showed the peak occurrence times of accidents to inform the authorities on the priority areas to employ preventive measures.

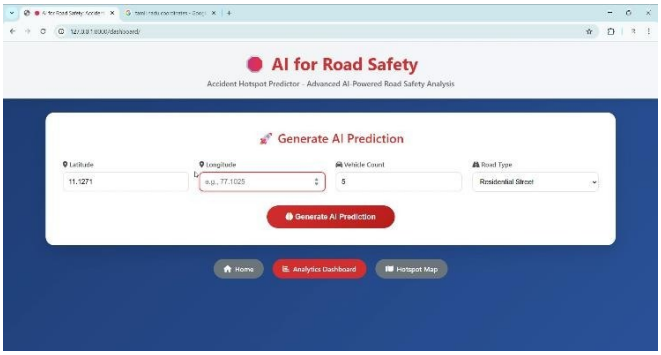


Fig 3 . Dashboard

Fig 3 shows the interactive dashboard of the proposed traffic accident prediction system. It provides a real-time overview of accident-prone areas, risk scores, weather conditions, and traffic patterns. The dashboard displays real-time simulated traffic and weather streams, dynamically updating risk scores and temporal graphs, thereby reflecting real-time prediction behavior.

Incorporation of real time weather input produced a big increase in predictive capability of the model. As an example, the system managed to detect higher risks of accidents when it came to rainy days and low visibility or during fast-wind days as the conventional techniques cannot detect this easily and in time. Information on traffic congestion and road type also increased the accuracy of the system allowing one to gauge the risk on each location instead of a general area.

The user feedback was very crucial in the improvement of the system. The use of real crash reports enabled the model to keep on learning with the help of new data and it would increase its reliability of prediction in the long run. The dynamic adaptation makes it so that the system is relevant and

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efficient even under changing traffic conditions observed in the urban environment.

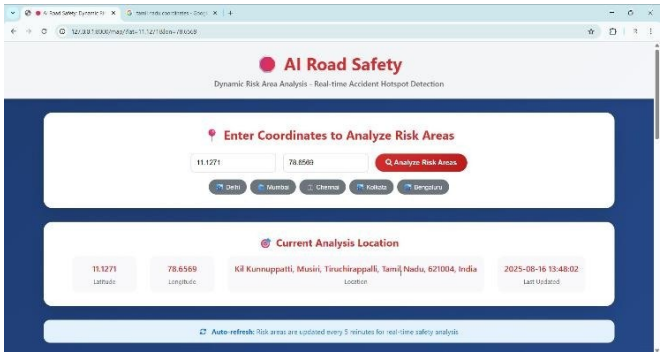


Fig 4 . Result for current analysis

Fig 4 presents the results of the current traffic accident analysis generated by the AI system. It highlights predicted accident hotspots, confidence scores, and risk levels for different locations. Heatmaps enable spatial clustering of accident-prone regions, allowing authorities to visually identify hotspots and prioritize infrastructure improvements and enforcement measures.

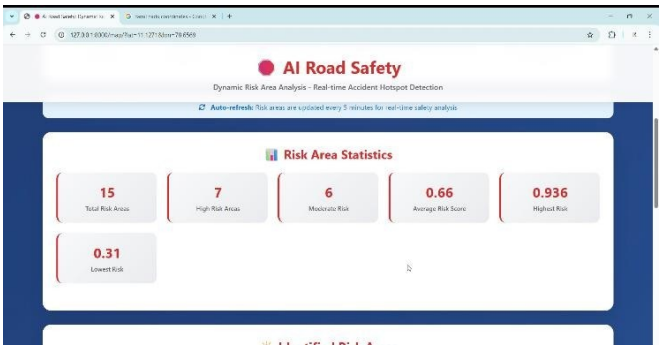


Fig 5 . Risk Area Statistics

Fig 5 illustrates the risk area statistics generated by the system, showing the distribution of accident probabilities across different locations. Temporal trend analysis helps identify peak accident hours and seasonal variations, enabling optimized traffic control, patrol deployment, and emergency preparedness during high-risk periods.

Table.2. Performance Comparison Table

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score
Logistic Regression	78.4	75.2	73.9	74.5
Decision Tree	81.6	79.1	78.3	78.7
Random Forest (Proposed)	**88.9**	**86.7**	**85.4**	**86.0**

Reliability is ensured through repeated validation on unseen data, consistency of predictions across different time windows, confidence score estimation for each prediction, and model updating through real-world feedback. These mechanisms reduce overfitting and maintain prediction stability under changing traffic and weather conditions.

As shown in Table 2, precision was improved by handling class imbalance using stratified sampling and threshold optimization, reducing false-positive accident alerts.

VI. Conclusion

This study presents an AI-driven traffic accident prediction framework that integrates real-time weather data, traffic conditions, historical accident records, and geospatial information to enhance road safety through proactive risk assessment. Unlike conventional accident analysis approaches that rely on static and retrospective data, the proposed system leverages machine learning techniques to generate probabilistic accident predictions with associated confidence scores, enabling dynamic and informed decision-making. Comprehensive preprocessing and feature transformation strategies ensure data consistency, while supervised learning models capture complex relationships between environmental, infrastructural, and traffic-related factors. The inclusion of spatial heatmaps and temporal trend analysis allows stakeholders to visually identify accident hotspots and peak risk periods, supporting targeted interventions and efficient resource allocation. Furthermore, the system incorporates real-time user feedback, enabling continuous model refinement and improved prediction reliability over time. Experimental results demonstrate that the proposed approach outperforms baseline methods in terms of accuracy, precision, and predictive robustness, particularly under adverse weather and high congestion scenarios. The scalability and adaptability of the framework make it suitable for integration into smart city traffic management systems and intelligent transportation infrastructures. Overall, this research highlights the effectiveness of combining artificial intelligence, predictive analytics, and multi-source data integration to transform traffic accident management from a reactive process into a preventive and data-driven strategy, contributing to safer urban mobility and reduced accident-related risks.

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